PROXIMAL POLICY OPTIMIZATION

INTERMEDIATE DEEP LEARNING

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PREVIOUSLY…

ACTOR-CRITIC – ADVANTAGE ACTOR-CRITIC (A2C)

PREVIOUSLY …

ACTOR-CRITIC

- Actor-critic combines policy-based and value-based methods
- Uses the advantage function to replace raw reward signals from the environment

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$

- In practice, we approximate V_π and estimate A_π from it
  - Using n-step return:

$$A_\pi^{\text{n-step}}(s_t,a_t) \approx r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_\pi(s_{t+n}) - V_\pi(s_t)$$

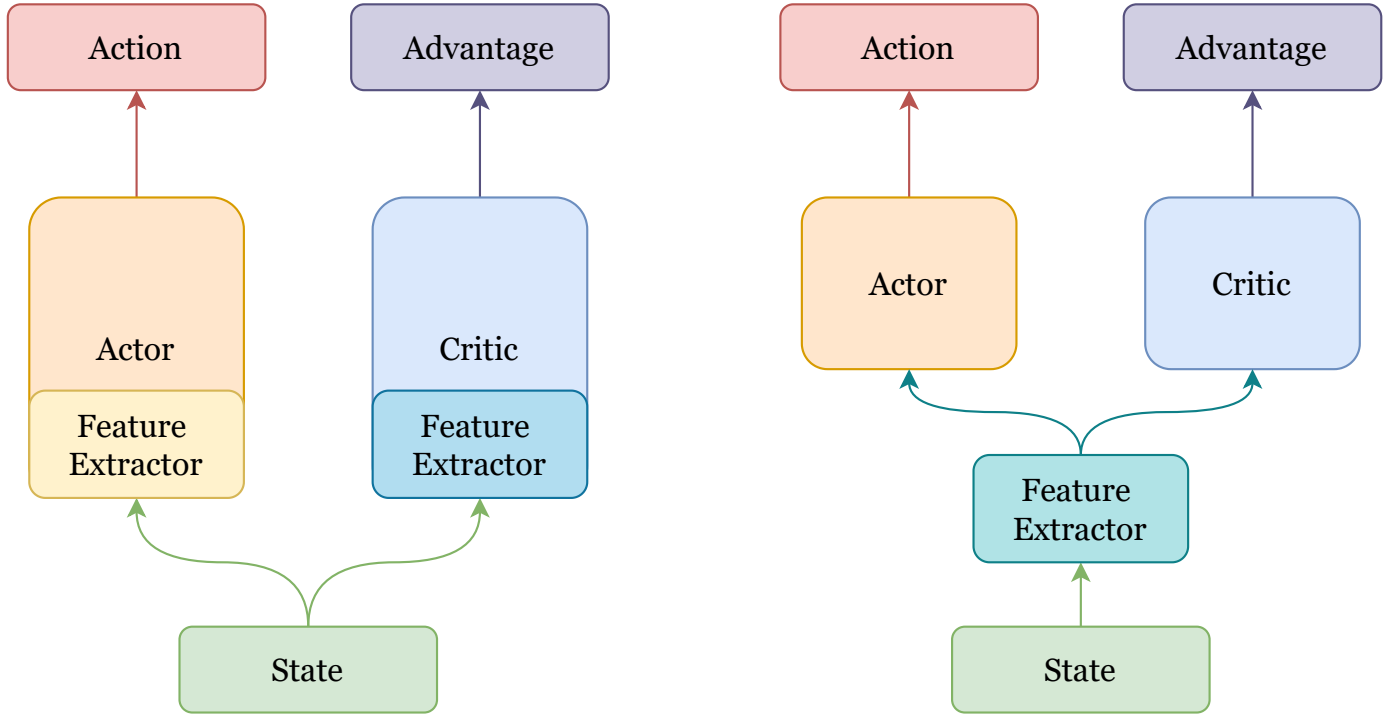
  - Or using Generalized Advantage Estimation (GAE) :

$$A_\pi^{\text{GAE}}(s_t,a_t) = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

PREVIOUSLY …

ADVANTAGE ACTOR-CRITIC (A2C)

- Entropy regularization encourages exploration
  - High entropy → more random actions → better exploration
  - Low entropy → more deterministic policy → less exploration
- Actor and critic often share the same feature extractor



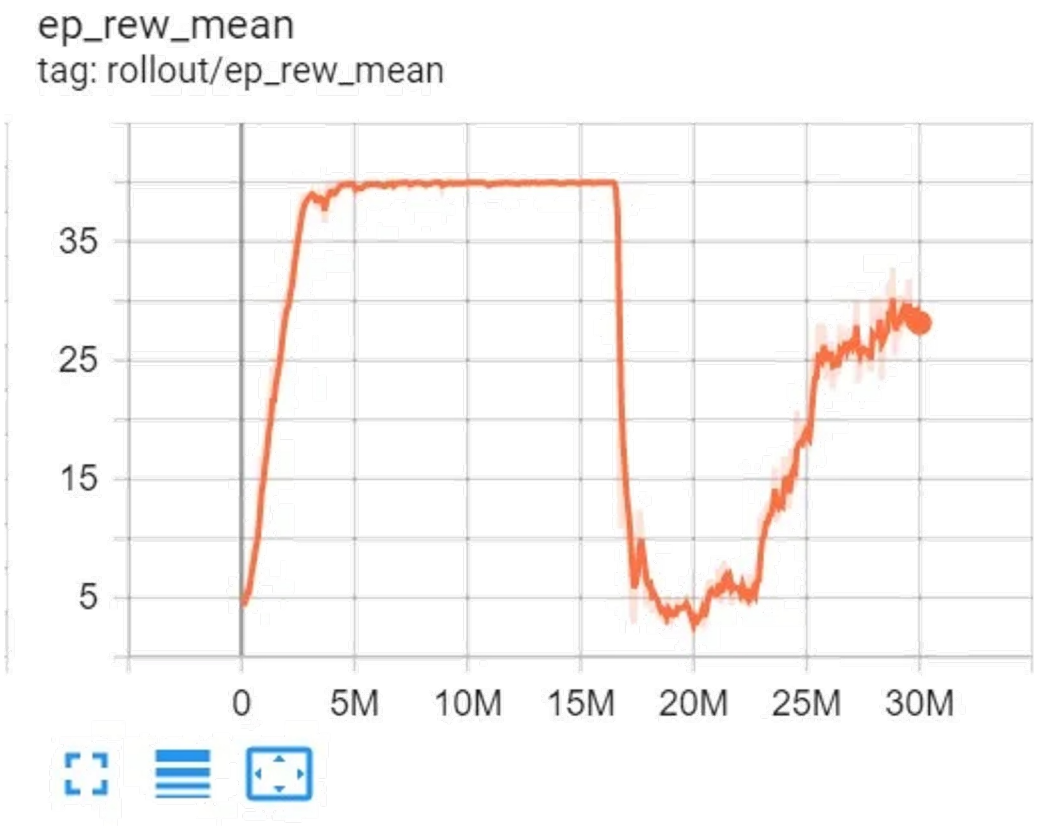
INTRODUCTION

PERFORMANCE COLLAPSE – SURROGATE POLICY GRADIENT

INTRODUCTION

PERFORMANCE COLLAPSE

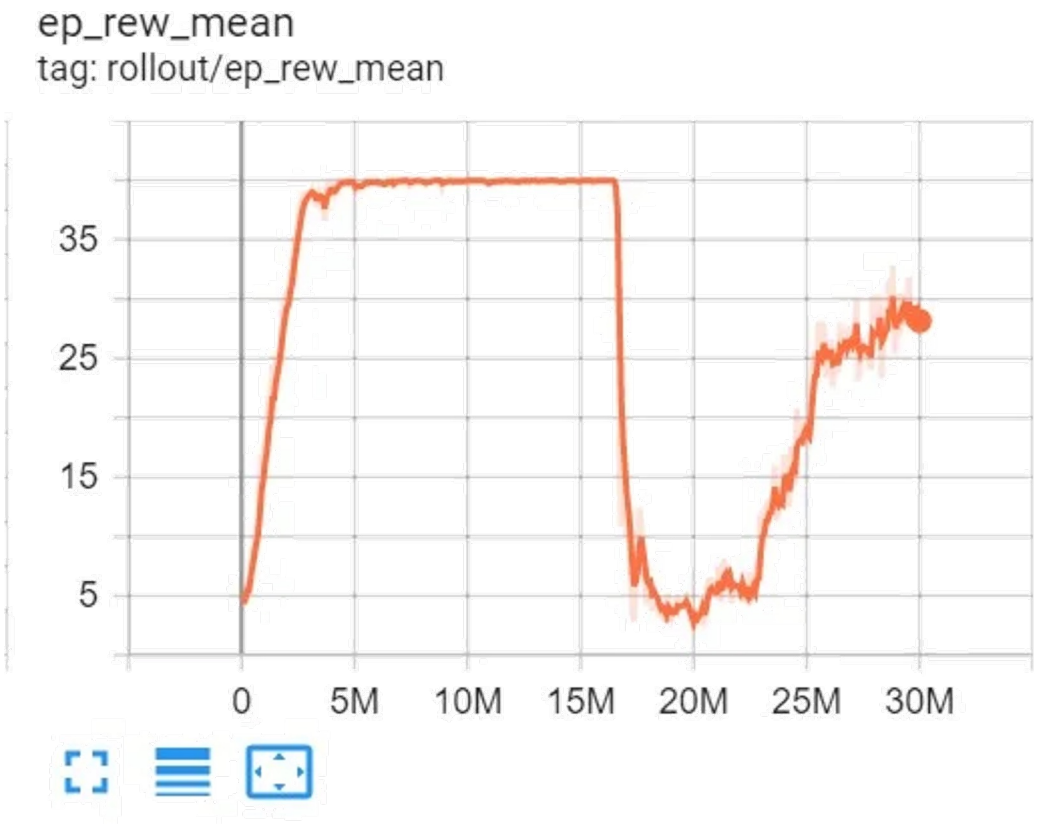
- Instability is one of the biggest enemies of deep reinforcement learning
- In policy gradient methods, we sometimes observe a performance collapse during training 👉
- Possible cause?



INTRODUCTION

PERFORMANCE COLLAPSE

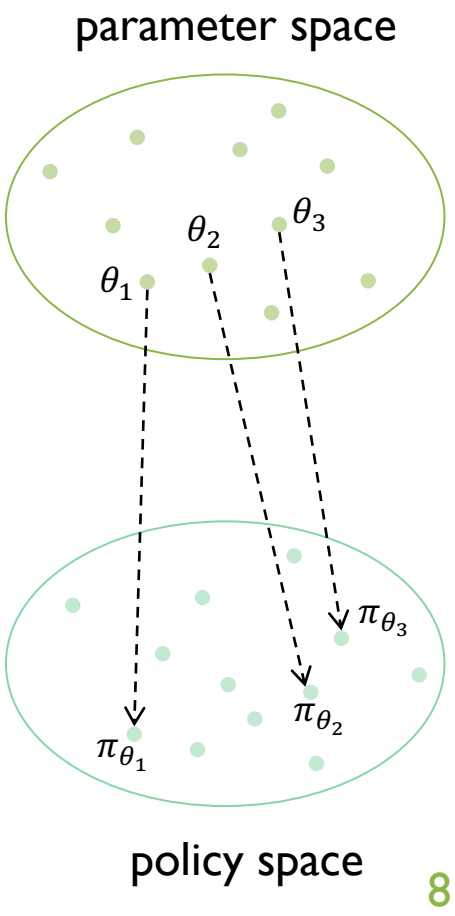
- Possible cause: the policy π_θ changes too abruptly, moving too far from the previous policy
- We already take measures to prevent large, unstable updates:
- α helps limit update size
- In last TD we also used gradient clipping
- However, the problem persists…



INTRODUCTION

PERFORMANCE COLLAPSE

- Policy Space ≠ Parameter Space
- We optimize in the parameter space (θ),
- but have no direct control over the policy space (π_θ)
- A small change in parameters does not always mean a small change in policy
- $d_\theta(\theta_1, \theta_2) = d_\theta(\theta_2, \theta_3) \nLeftrightarrow d_{\pi_\theta}(\pi_{\theta_1}, \pi_{\theta_2}) = d_{\pi_\theta}(\pi_{\theta_2}, \pi_{\theta_3})$
- $\rightarrow$ can lead to unstable updates



INTRODUCTION

SURROGATE POLICY GRADIENT

- Since the problem lies in the step size, our intuition:
- Limit policy updates so that performance does not degrade
- We need to measure the performance difference between two policies

INTRODUCTION

SURROGATE POLICY GRADIENT

- A natural way to compare two policies is by their expected returns:

$$J(\pi) = E_{\tau\sim\pi}[R(\tau)]$$

- The performance difference is:

$$J(\pi')-J(\pi)$$

- And then can define a relative policy performance identity as:

$$J(\pi')-J(\pi) = E_{\tau\sim\pi'}\left[\sum_{t=0}^{T}\gamma^t A_\pi(s_t,a_t)\right]$$

- Intuitively, if the new policy π' performs better than the old policy π, then it should have a positive expected advantage under π.

SURROGATE POLICY GRADIENT

- We can prove that the definition is correct:

$$E_{\tau\sim\pi'}\left[\sum_{t=0}^{T}\gamma^t A_\pi(s_t,a_t)\right] = E_{\tau\sim\pi'}\left[\sum_{t=0}^{T}\gamma^t\left(r_t + \gamma V_\pi(s_{t+1}) - V_\pi(s_t)\right)\right]$$

- By regrouping terms:

$$= E_{\tau\sim\pi'}\left[\sum_{t=0}^{T}\gamma^t r_t\right] + E_{\tau\sim\pi'}\left[\sum_{t=0}^{T}\gamma^{t+1}V_\pi(s_{t+1}) - \sum_{t=0}^{T}\gamma^t V_\pi(s_t)\right]$$

- Simplifying:

$$= J(\pi') - E_{\tau\sim\pi'}[V_\pi(s_0)] = J(\pi') - E_{\tau\sim\pi'}[J(\pi)]$$

- We obtain:

$$J(\pi') - J(\pi) = E_{\tau\sim\pi'}\left[\sum_{t=0}^{T}\gamma^t A_\pi(s_t,a_t)\right]$$

INTRODUCTION

SURROGATE POLICY GRADIENT

$$J(\pi') - J(\pi) = E_{\tau \sim \pi'} \left[\sum_{t=0}^{T} \gamma^t A_\pi(s_t, a_t) \right]$$

- Since this term also quantifies the improvement made by updating the policy, can we use it as an objective function for the policy update?

INTRODUCTION

SURROGATE POLICY GRADIENT

$$J(\pi') - J(\pi) = E_{\tau \sim \pi'} \left[\sum_{t=0}^{T} \gamma^t A_\pi(s_t, a_t) \right]$$

- Since this term also quantifies the improvement made by updating the policy, can we use it as an objective function for the policy update?
- Unfortunately, no 😞
  - It depends on trajectories sampled using the new policy π' , which are only available after the update

SURROGATE POLICY GRADIENT

- To remedy this paradox, we find a way to alter it to use the old policy π
- We assume that successive policies π and π' are **relatively close**, so that the state distributions from them are similar
- Then we can approximate $J(\pi')-J(\pi)$ by trajectory from the old policy:

$$J(\pi') - J(\pi) \approx E_{\tau\sim\pi}\left[\sum_{t=0}^{T}\gamma^t A_\pi(s_t,a_t)\frac{\pi'(a_t \mid s_t)}{\pi(a_t \mid s_t)}\right] = J_\pi^{\mathrm{CPI}}(\pi')$$

- CPI stands for Conservative Policy Iteration

SURROGATE POLICY GRADIENT

- We verify that optimization under objective $J_{\pi}^{\mathrm{CPI}}(\pi')$ is still doing policy gradient ascent, which means

$$\nabla_{\theta} J_{\pi_{\bar{\theta}}}^{\mathrm{CPI}}(\pi_{\theta}) = \nabla_{\theta} J(\pi_{\theta})$$

- Where $\pi' = \pi_{\theta}$ and $\pi = \pi_{\bar{\theta}}$(old policy)

SURROGATE POLICY GRADIENT

- First:

$$\nabla_\theta J^{\text{CPI}}_{\pi_{\bar{\theta}}}(\pi_\theta) = \nabla_{\theta E_{\tau\sim\pi_{\bar{\theta}}}}\left[\sum_{t=0}^{T}\gamma^t A_{\pi_{\bar{\theta}}}(s_t,a_t)\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\bar{\theta}}(a_t \mid s_t)}\right]$$

- Since only the numerator depends on θ, we can write:

$$= E_{\tau\sim\pi_{\bar{\theta}}}\left[\sum_{t=0}^{T}\gamma^t A_{\pi_{\bar{\theta}}}(s_t,a_t)\frac{\nabla_\theta\pi_\theta(a_t \mid s_t)}{\pi_{\bar{\theta}}(a_t \mid s_t)}\right]$$

- At the beginning of training, $\bar{\theta} = \theta$:

$$= E_{\tau\sim\pi_{\bar{\theta}}}\left[\sum_{t=0}^{T}\gamma^t A_{\pi_{\bar{\theta}}}(s_t,a_t)\frac{\nabla_\theta\pi_\theta(a_t \mid s_t)}{\pi_\theta(a_t \mid s_t)}\right]$$

- With log derivative trick:

$$= E_{\tau\sim\pi_{\bar{\theta}}}\left[\sum_{t=0}^{T}\gamma^t A_{\pi_{\bar{\theta}}}(s_t,a_t)\nabla_\theta \log\pi_\theta(a_t \mid s_t)\right] = \nabla_\theta J(\pi_\theta)$$

MONOTONIC IMPROVEMENT THEORY

- Going back to the beginning, our goal is for each policy update to improve performance, that is, $J(\pi') - J(\pi) \geq 0$
- However, using the surrogate objective $J_{\pi}^{\mathrm{CPI}}(\pi')$ to approximate this difference introduces an approximation error
- Therefore, we need to better understand this error

INTRODUCTION

MONOTONIC IMPROVEMENT THEORY

- When π and π' are sufficiently close (measured by KL divergence), then we can write a relative policy performance bound:

$$\left|\left(J(\pi') - J(\pi)\right) - J_{\pi}^{\mathrm{CPI}}(\pi')\right| \leq C\sqrt{E_t\left[\mathrm{KL}\left(\pi'(a_t \mid s_t)||\pi(a_t \mid s_t)\right)\right]}$$

- Where C is a constant depends on γ and maximum reward, and KL is the KL divergence

INTRODUCTION

MONOTONIC IMPROVEMENT THEORY

- Reminder: Kullback–Leibler (KL) Divergence
- A measure of how one probability distribution P diverges from a reference distribution Q:

$$\mathrm{KL}(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}$$

INTRODUCTION

MONOTONIC IMPROVEMENT THEORY

$$J(\pi') - J(\pi) \geq J_{\pi}^{\mathrm{CPI}}(\pi') - C\sqrt{E_t\left[\mathrm{KL}\left(\pi'(a_t \mid s_t)||\pi(a_t \mid s_t)\right)\right]}$$

- The bound allows us to guarantee non-negative improvement by choosing an update whose lower bound is non-negative.
- How?

INTRODUCTION

MONOTONIC IMPROVEMENT THEORY

$$J(\pi') - J(\pi) \geq J_{\pi}^{\mathrm{CPI}}(\pi') - C\sqrt{E_t\left[\mathrm{KL}\left(\pi'(a_t \mid s_t)||\pi(a_t \mid s_t)\right)\right]}$$

- The bound allows us to guarantee non-negative improvement by choosing an update whose lower bound is non-negative.
- If $\pi' = \pi$
  - $E_t[A_\pi(s_t, a_t)] = 0 \Rightarrow J_{\pi}^{\mathrm{CPI}}(\pi') = 0,$
  - $\mathrm{KL}\left(\pi'(a_t \mid s_t)||\pi(a_t \mid s_t)\right) = 0.$
  - Hence $J(\pi') - J(\pi) \geq 0$

INTRODUCTION

MONOTONIC IMPROVEMENT THEORY

$$J(\pi') - J(\pi) \geq J_{\pi}^{\mathrm{CPI}}(\pi') - C\sqrt{E_t\left[\mathrm{KL}\left(\pi'(a_t \mid s_t)||\pi(a_t \mid s_t)\right)\right]}$$

- This is the Monotonic Improvement Theory that provides a theoretical guarantee:
  - A new policy π' will not perform worse than the old policy π, as long as the policy change is small enough.
- Our optimization problem becomes:

$$\arg\max_{\pi'}\left(J_{\pi}^{\mathrm{CPI}}(\pi') - C\sqrt{E_t\left[\mathrm{KL}\left(\pi'(a_t \mid s_t)||\pi(a_t \mid s_t)\right)\right]}\right)$$

MONOTONIC IMPROVEMENT THEORY

- To implement this constraint on the policy change, a simple idea is to set a hard bound on the KL divergence:

$$E_t\left[\mathrm{KL}\left(\pi'(a_t \mid s_t)||\pi(a_t \mid s_t)\right)\right] \leq \delta$$

- which only allows new policies within a small neighborhood (trust region) around the current one to be considered

MONOTONIC IMPROVEMENT THEORY

- The resulting optimization problem becomes:

$$\max_{\pi'}\left(J_{\pi}^{\mathrm{CPI}}(\pi')\right) \rightarrow \max_{\theta} E_t\left[\frac{\pi_\theta(a_t|s_t)}{\pi_{\bar{\theta}}(a_t|s_t)} A_{\pi_{\bar{\theta}}}(a_t|s_t)\right]$$

  - Subject to

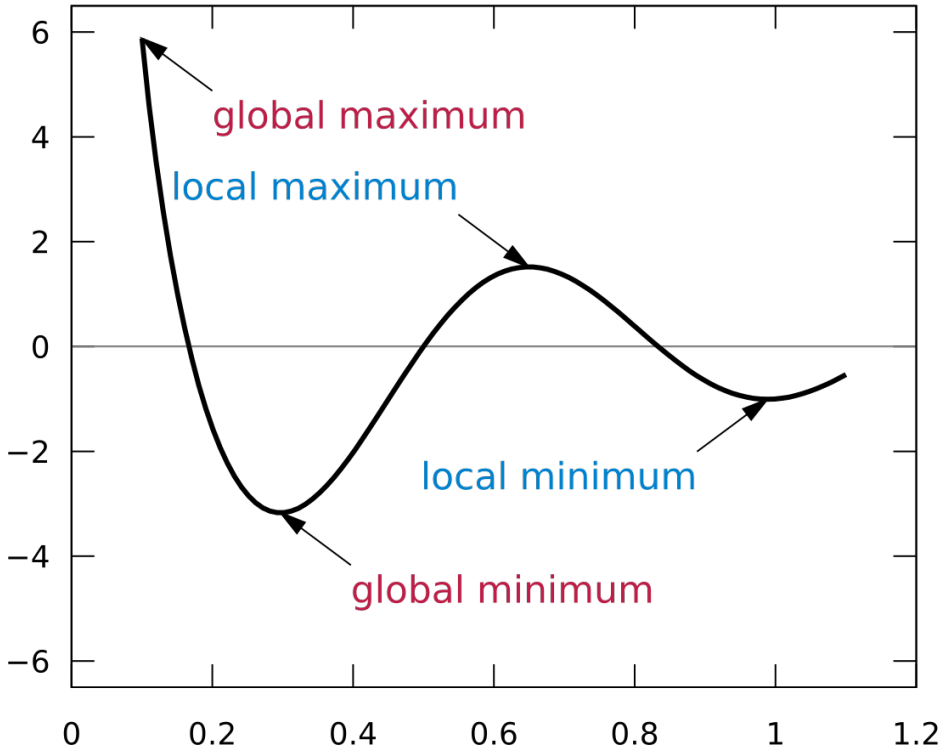
$$E_t\left[\mathrm{KL}\left(\pi_\theta(a_t \mid s_t)||\pi_{\bar{\theta}}(a_t \mid s_t)\right)\right] \leq \delta$$

- Algorithms such as Trust Region Policy Optimization (TRPO) solve this problem

INTRODUCTION

MONOTONIC IMPROVEMENT THEORY

- Remark:
- The monotonic improvement theory does not guarantee convergence to the optimal policy.
- E.g. the algorithm stuck at a local maximum and no longer change the policy



PROXIMAL POLICY OPTIMIZATION (PPO)

CLIPPED SURROGATE OBJECTIVE – MULTIPLE PASS

PROXIMAL POLICY OPTIMIZATION (PPO)

CLIPPED SURROGATE OBJECTIVE

$$E_t\left[\mathrm{KL}\left(\pi_\theta(a_t \mid s_t)||\pi_{\bar{\theta}}(a_t \mid s_t)\right)\right] \leq \delta$$

- Enforcing such a hard constraint is computationally expensive.
  - We must compute the new policy, then perform a line search to ensure it satisfies the constraint.

PROXIMAL POLICY OPTIMIZATION (PPO)

CLIPPED SURROGATE OBJECTIVE

- Proximal Policy Optimization (PPO) 🧙‍♀️ to the rescue!
  - Easy to implement
  - Computationally efficient
  - Very popular in practice 👑

PROXIMAL POLICY OPTIMIZATION (PPO)

CLIPPED SURROGATE OBJECTIVE

- Main idea: limiting each policy update through a clipped surrogate objective, preventing overly large policy changes while maintaining efficient learning

$$J^{\text{CLIP}}(\theta) = E_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)]$$ 🤔

- Where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\overline{\theta}}(a_t|s_t)}$$

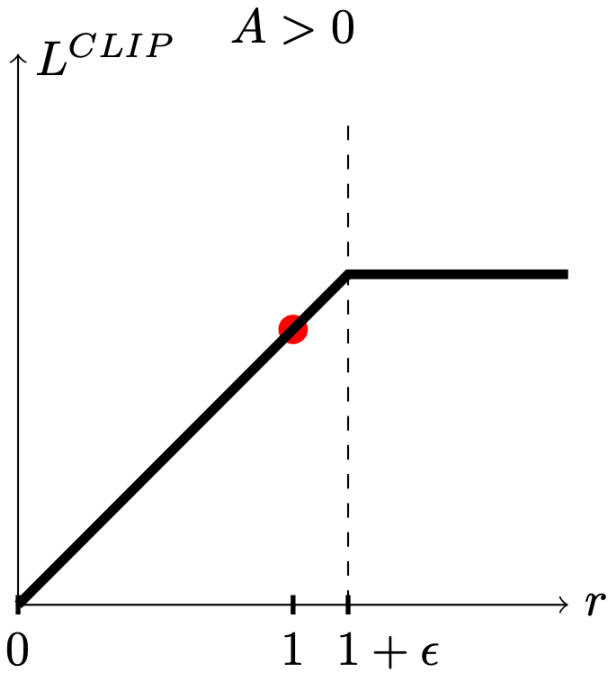
- Compare to CPI:

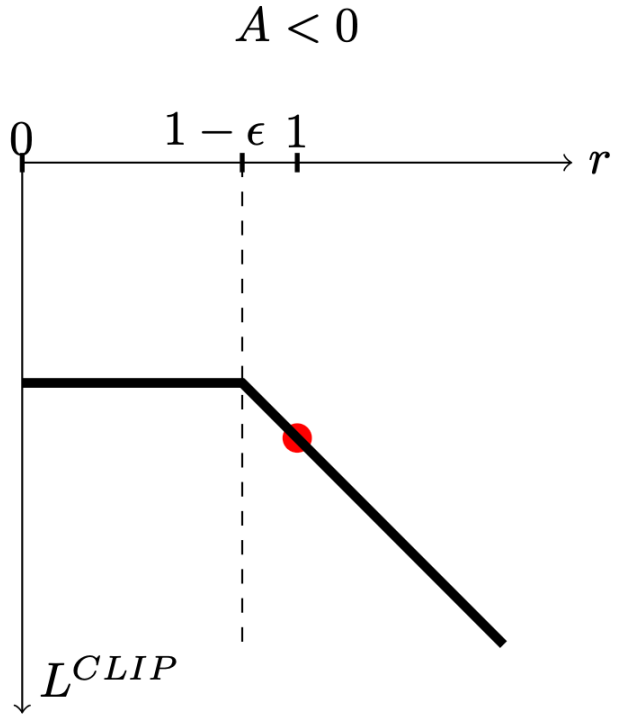
$$J^{\text{CPI}}(\theta) = E_t[r_t(\theta)A_t]$$

PROXIMAL POLICY OPTIMIZATION (PPO)

CLIPPED SURROGATE OBJECTIVE

$$J^{\text{CLIP}}(\theta) = E_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)]$$





PROXIMAL POLICY OPTIMIZATION (PPO)

MULTIPLE PASS

- Since PPO assume that samples are generated from $\pi_{\bar{\theta}}$, the data distribution does not depend on the current policy during the update → This allows us to reuse the same trajectories multiple times for parameter updates
  - This contrasts with A2C, which typically uses each sample only once.
- However, because the objective still depends on $\pi_{\bar{\theta}}$, PPO remains on-policy
  - Once $\pi_{\bar{\theta}}$ changes, the old experience must be discarded

PROXIMAL POLICY OPTIMIZATION (PPO)

REMARKS

- PPO actually makes a strong assumption about the smallness of each policy update
  - The new policy π_θ is assumed to stay close to the old one $\pi_{\overline{\theta}}$
- In practice, this assumption is not always strictly satisfied…
- Yet, like magic, PPO still works remarkably well without a formal theoretical guarantee 😳
- This empirical robustness is largely due to
  - Using a small learning rate, which keeps updates moderate
  - The clipping mechanism, which limits excessive policy changes
  - Multiple epochs with mini-batches, improving sample efficiency while maintaining stability
  - The entropy bonus, which prevents premature convergence to deterministic policies

CODING

TD6: PPO